

Muhammad Aleem Azhar

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Computer Science senior at Rhodes College (Statistics minor, May 2026) focused on agentic AI, applied machine learning, and data-intensive backend systems. I build evaluation-first pipelines, LLM-powered workflows, and evidence-grounded systems that reason over context, handle noisy data, and turn model outputs into reliable, customer-useful decisions.

EDUCATION

Rhodes College

Expected May 2026

B.S. in Computer Science, Minor in Statistics

- Selected coursework: Algorithms, Databases, Machine Learning, Regression Analysis, Stochastic Processes, Linear Algebra, Probability & Statistics, Software Engineering
- Study Abroad: Yonsei University, Seoul - Artificial Intelligence and Computer Vision (Aug 2024 - Dec 2024)

TECHNICAL SKILLS

ML/NLP: PyTorch, scikit-learn, TensorFlow, OpenCV, LLM workflows, RAG concepts, evaluation, prompt testing, ablations, retrieval/ranking

Data: Python, pandas, NumPy, SQL, PostgreSQL, TimescaleDB, ETL, telemetry, schema validation, dashboards, experimental analysis

Systems: FastAPI, Flask, REST APIs, Redis, Docker, GitHub Actions, Linux, CI/CD, testing, observability, reproducible runs

RESEARCH & EXPERIENCE

Researcher, Generative Residual Learning for Noise Resilient Time Series Forecasting

Sep 2025 - Present

Rhodes College

Machine Learning Research

- Developing a Regression Wasserstein GAN (RWGAN) framework that separates deterministic temporal signals from stochastic residual modeling to support safer forecasting under noise, volatility, and heavy-tailed disturbances.
- Built reproducible ML experiments with chronological splits across energy, air-quality, NASDAQ, and Bitcoin datasets; benchmarked against ARIMA, Random Forest, MLP, LSTM, and GAN baselines.
- Established error bounds linking forecast quality to regression accuracy and residual-distribution mismatch; evaluated reliability with RMSE, MAE, error variance, MAPE, and robustness-focused diagnostics.

Teaching Assistant, Fundamentals of Computer Science

Jun 2025 - Aug 2025

Johns Hopkins University, Center for Talented Youth

Computer Science Education

- Taught Python fundamentals, data structures, and introductory ML concepts; improved comprehension **40%** and engagement **35%** across multi-campus delivery.
- Led labs and 1:1 mentoring for **30+** students, emphasizing decomposition, debugging, API-style thinking, and testable code; created guides that improved lesson delivery efficiency **30%**.

Data Manager & Administrative Support

Jan 2023 - Present

Rhodes College Financial Aid Office

Data Operations

- Automated reporting and reconciliation workflows with Python, Excel, and SQL validation, reducing cycle time **30%** while improving data accuracy, controls, and audit readiness across student-employment operations.

RHOK-SAT CubeSat Research Intern

May 2023 - Jul 2023

Rhodes College

Telemetry Systems

- Built a Python telemetry pipeline and Plotly/Dash dashboard, reducing manual analysis time **40%**; surfaced packet corruption and loss explicitly to support reliable monitoring and downstream analysis.

SELECTED PROJECTS

City Pulse

Apr 2025 - Aug 2025

Transformer-based sentiment and emotion analytics over time-stamped social text

- Ingested short posts, inferred timestamps from location metadata, applied transformer-based sentiment and emotion models, and indexed outputs for fast district- and duration-based API queries.
- Analyzed robustness failures from slang, sarcasm, and thread drift, motivating evaluation checks on whether small context changes alter model claims or customer-facing recommendations.

AI Social Mirror

May 2025 - Jun 2025

Evaluation-first LLM writing-feedback system with behavior monitoring

- Tracked behavior under small edits, logged regressions, and prioritized feedback users could inspect or contest, treating LLM evaluation as a continuous production property rather than a one-time report.

BiteQuest

Jan 2026 - Present

Personalized ranking and dish-intelligence platform for restaurant discovery

- Designed a mobile-first recommendation architecture using Next.js, FastAPI, PostgreSQL, Redis, and explainable trust scoring to combine restaurant signals, dish extraction, personalization, and cached ranking.